

PVCELL–25 power cell

Rev C0 | 2026–10–04 | Schematics authored by Chinmoy Bhuyan | 25 kW (27.5 kW max) bidirectional 4–switch buck–boost cell, 2 x 4 x SG2M040170HJ

01 CELL connector, logic supply, ID

40–way CELL contract to CTRL–C2000, +24V = safety–switched +24V_GD;
LMR36015 3.3 V + 5 V (driver logic side), TPS3710 on RDY, ID 4.99k, AGND tie

File: sheets/01_cell.kicad_sch

02 Fail–safe RS–422, fault latch

AM26LV32E receivers with external fail–safe bias (open = gate off),
FLT driver, OC/OT fault latch, EN_DRV, PWM gating

File: sheets/02_logic.kicad_sch

03 Power stage, decoupling, dampers

Leg A: S1/S2 on port A, leg B: S3/S4 on port B, 2 x SG2M040170HJ each;
film decoupling and RC damper per leg at the device pins

File: sheets/03_power.kicad_sch

04 Film banks, bleeders, inductor

Port A/B C4AQ banks + TNPV bleeders, bus studs to PV–PORT,
off–board CUSTOM inductor on studs SW_A / SW_B, winding NTC header

File: sheets/04_bus.kicad_sch

05 Inductor current, temperatures, trips

AN1 = LEM LA 150–P + driver, AN2/AN3 = heatsink/winding temperature,
OC window + heatsink OT comparators, 2.5 V reference, NTC pass–through

File: sheets/05_sense.kicad_sch

06 Gate drive S1 (leg A)

gdrv.channel() rev 5: NSI6651ASC–Q1 + UCC14241–Q1 +18/–3.5 V,
3.75/2.5 ohm + Kelvin R per device, A03400A clamps,
SC booster 22R, interlock with PWM2, DESAT to the drain

File: sheets/06_gd_s1.kicad_sch

07 Gate drive S2 (leg A)

gdrv.channel() rev 5: NSI6651ASC–Q1 + UCC14241–Q1 +18/–3.5 V,
3.75/2.5 ohm + Kelvin R per device, A03400A clamps,
SC booster 22R, interlock with PWM1, DESAT to the drain

File: sheets/07_gd_s2.kicad_sch

08 Gate drive S3 (leg B)

gdrv.channel() rev 5: NSI6651ASC–Q1 + UCC14241–Q1 +18/–3.5 V,
3.75/2.5 ohm + Kelvin R per device, A03400A clamps,
SC booster 22R, interlock with PWM4, DESAT to the drain

File: sheets/08_gd_s3.kicad_sch

09 Gate drive S4 (leg B)

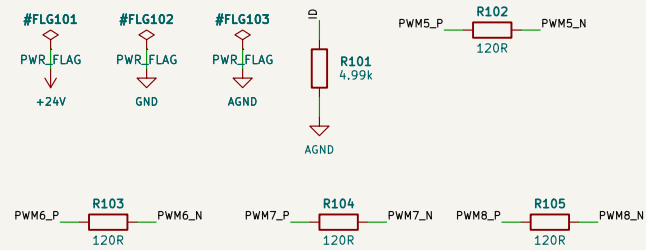
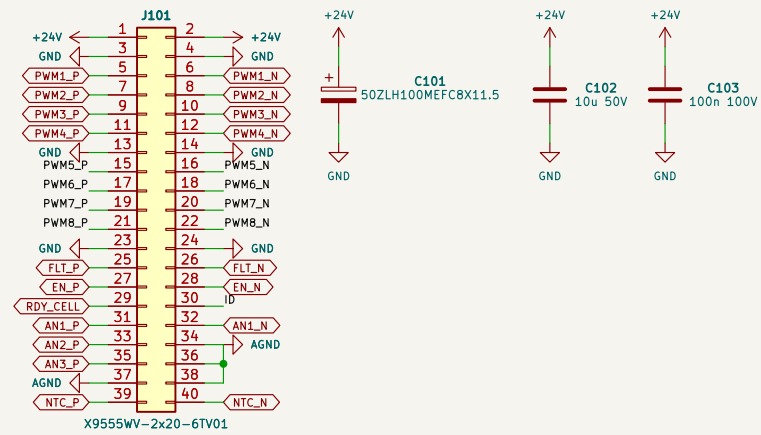
gdrv.channel() rev 5: NSI6651ASC–Q1 + UCC14241–Q1 +18/–3.5 V,
3.75/2.5 ohm + Kelvin R per device, A03400A clamps,
SC booster 22R, interlock with PWM3, DESAT to the drain

File: sheets/09_gd_s4.kicad_sch

CALCULATED, not measured: design_check() asserts every number on each build (outputs/PVCELL–25_design_check.txt); component values are read from sim/out/pv_design/cell_spec.json.
Local trips: inductor OC +/-72.2 A (68.5–76.0 A), gates off 0.94 us (PWM) / 1.73 us (EN) vs 3.66 us allowed; heatsink OT 95.0 C (92.3–97.9 C), open probe trips; DESAT per driver; CTRL backup peak 106.0 A.
AN1 = LEM LA 150–P on inductor lead A: 20.0 mV/A differential, CM 1.25 V, < 1 ohm, linear +/-114 A, 0.52 us, 0.0759 A/LSB at the CTRL. Bus OV: no local comparator (CTRL path 26 us -> 1115 V, 0.835 x V_DSS).
Fail–safe off: open, unplugged, unpowered or tri–stated pairs read <= –0.29 V = LOW; RDY push–pull; +24V removal drops RDY at >= 19.7 V; +3V3 or +5V loss -> drivers off (VCC1 UVLO / EN low). No bootstrap: 100 % duty OK.
Domains: PELV, HV, GD_S1..S4, PE. Isolators: NSI6651ASC–Q1 + UCC14241–Q1 x 4, LA 150–P, L_CELL. CONFORMAL COATING to PD1 (IEC 60664–3) over every HV–PELV barrier part and the power stage (IC–17, IC–22). 1000 V DC (ECO–10).
LAYOUT: loop <= 20 nH, bank <= 15 nH from each leg, damper loop <= 3 nH, clamp branch <= 1.5 nH, Miller clamp loop <= 1 nH, drain–pin slot. BENCH: Miller margin 0.53 V, LA 150–P dv/dt, damper power, pad PD test.

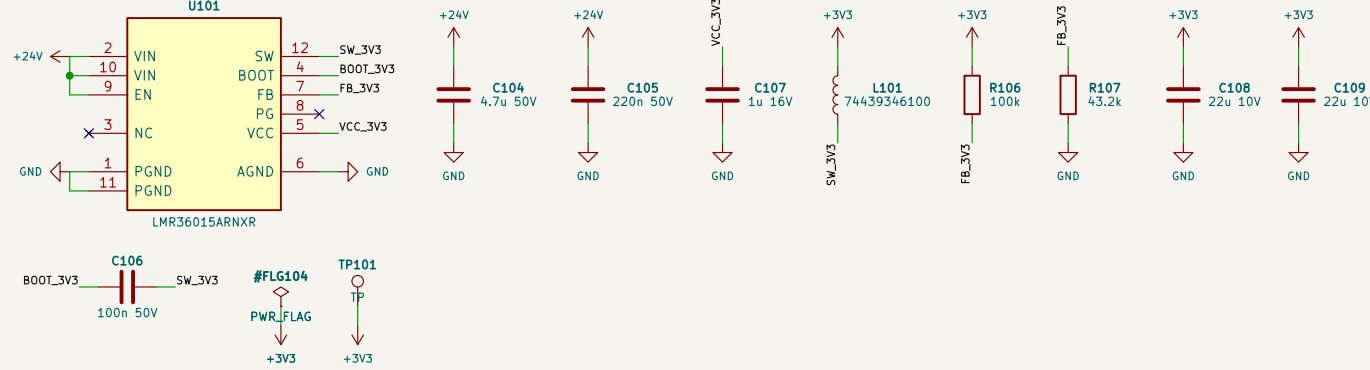
Not bench–validated. Values: sim/out/pv_design/cell_spec.json; checks: design_check() Sheet 1 Schematics authored by Chinmoy Bhuyan HV–PELV reinforced 1000 V DC (ECO–10); barriers listed on the cover		
Sheet: / File: PVCELL–25.kicad_sch		
Title: PVCELL–25 power cell – Cover / Hierarchy		
Size: A3	Date: 2026–10–04	Rev: C0
KiCad E.D.A. 10.0.6	Id: 1/10	

CELL CONNECTOR (INTERFACES.CELL)



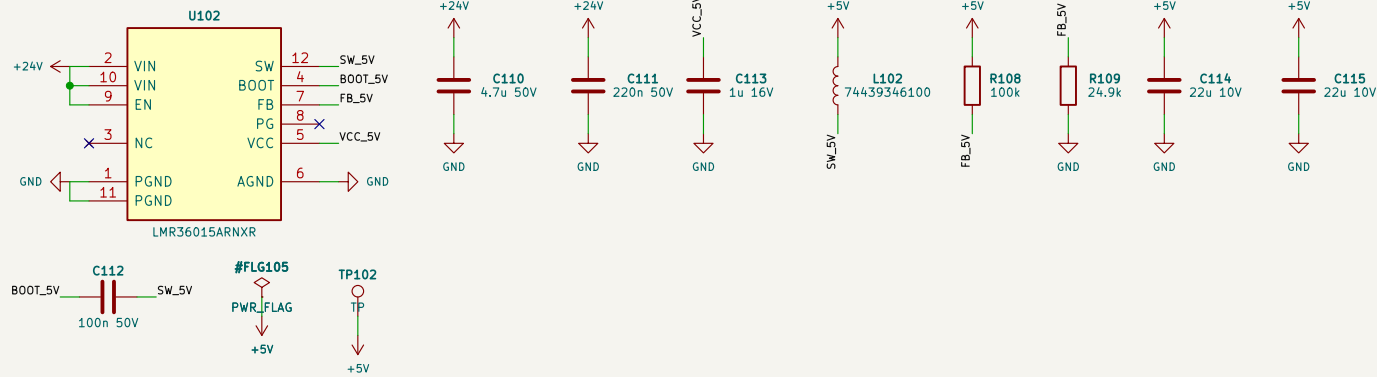
Ribbon 1:1 to a CTRL-C2000 cell port. AN2_N/AN3_N = AGND here (single-ended 0-2.5 V, AN_N tied at the source per the CTRL AFE). NTC_P/N float to the second heatsink NTC (biased on CTRL). PWM5-8 unused: 120R only. +24V: 0.52 A of 1.0 A worst case (design_check); <= 600 uF on the rail.

LOGIC SUPPLY +3V3 (LMR36015, 400 KHZ)



24 V -> 3.315 V (100k/43.2k, SNVSB49D 10.2, as PV-PORT). Runs down to 4.2 V input, so the logic keeps every gate disabled while +24V collapses. Load <= 119 mA (datasheet maxima).

LOGIC SUPPLY +5V (LMR36015, 400 KHZ)



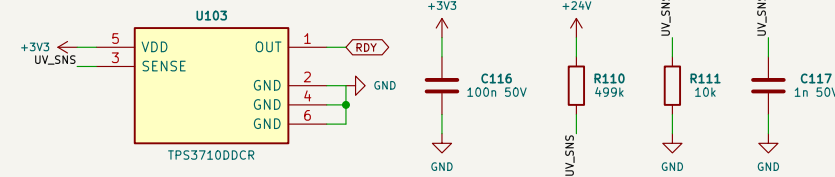
24 V -> 5.02 V (100k/24.9k); VCC1 of the four NSI6651 and the AHCT input stage - the driver input thresholds are specified at VCC1 = 5 V only (p7). Load <= 39 mA (datasheet maxima).

AGND - GND STAR POINT



Analog return (LA 150-P M / RM, +/-15 V COM, op amps, comparators, reference, NTCs, ID) meets GND only here.

+24V SUPERVISOR (TPS3710) ON RDY



OUT is open drain on the RDY line: RDY low below 19.66-20.40 V, released above 20.12-20.60 V. That is above the UCC14241-Q1 UVLO (17.1-18.9 V falling); EN_DRV drops and every driver pulls its gates to VEE while the bias is still regulating -> clean turn-off when the safety chain removes +24V_GD (D-012).

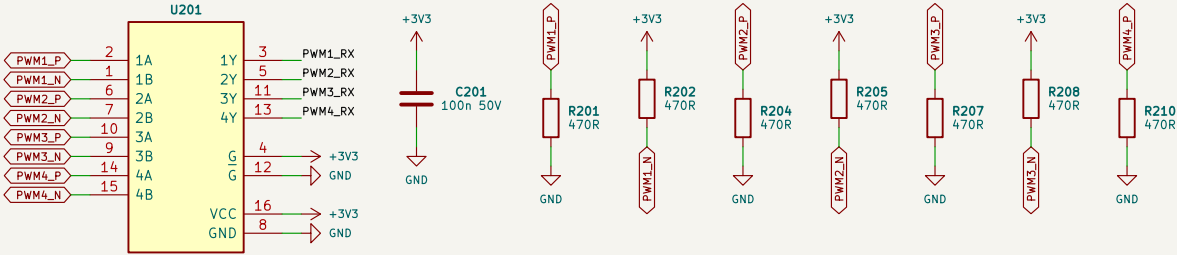
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Sheet 2
Schematics authored by Chinmoy Bhuyan
HV-PELV reinforced 1000 V DC (ECO-10); barriers listed on the cover

Sheet: /01 CELL connector, logic supply, ID/
File: 01_cell.kicad_sch

Title: PVCELL-25 power cell - CELL connector, logic supply, ID

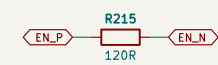
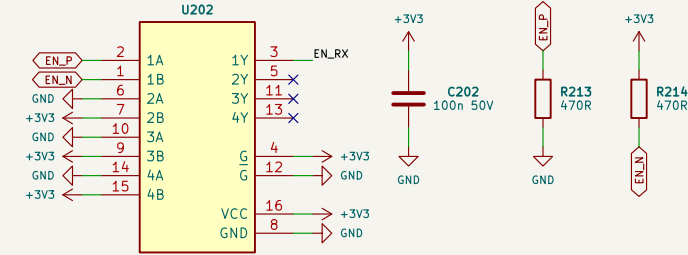
Size: A3 Date: 2026-10-04 Rev: C0
KiCad E.D.A. 10.0.6 Id: 2/10

PWM1-4 RECEIVERS (FAIL-SAFE)



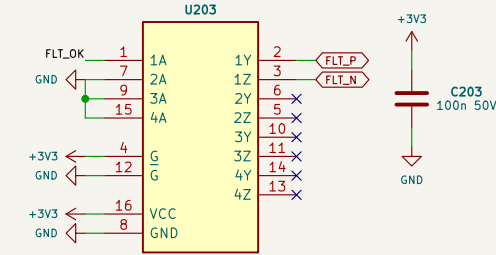
A = PWMx_P, B = PWMx_N, 470R P->GND, 470R N->+3V3, 120R across: an open, unplugged or unpowered pair reads -0.30 V / -0.29 V worst case (< V_{IT} -0.2 V) -> output LOW = gate off. The receiver's own open-input fail-safe (output HIGH) never applies: the bias always defines the pair. Driven high: 1.9 V.

EN RECEIVER (FAIL-SAFE)



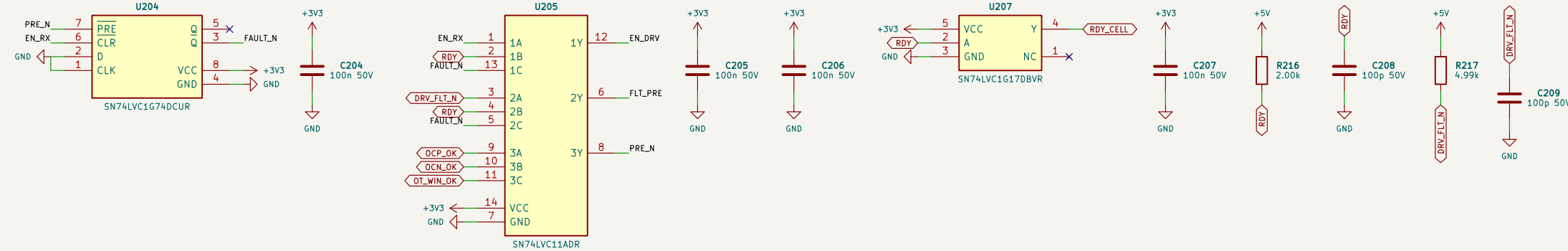
Same bias: open EN = LOW = drivers disabled. Spare receivers: A low, B high.

FLT DRIVER



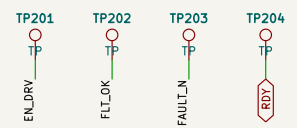
FLT_P = FLT_OK, FLT_N = /FLT_OK, CTRL receives the pair swapped with its own bias: healthy = -2 V = no fault; this board unpowered = AM26LV31E outputs HI-Z (SLSB4BC) -> CTRL bias -> fault.

FAULT LATCH, GATE ENABLE, FLT, RDY



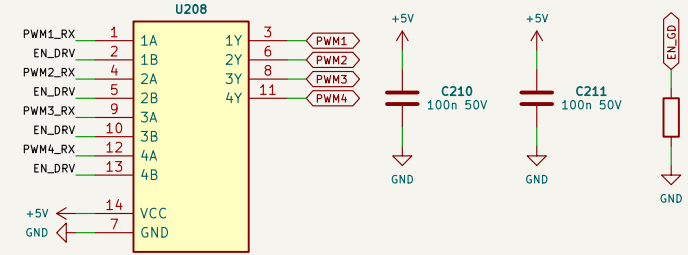
PRE_N = DCP_OK AND OCN_OK AND DT_WIN_OK (low on fault) sets the latch: FAULT_N low until EN drops with EN low AND a fault still present both /PRE and /FLR are low (Q = 0 = H, Table B-1), so FLT_OK = DRV_FLT_N AND RDY AND FAULT_N AND PRE_N reports the LIVE fault too (PVR-07 / INT-17): EN rising re-sets the latch. EN_DRV = EN_RX AND RDY AND FAULT_N drives every RST/EN, RDY (wired-AND: drivers, bias P₀, +24V and +15V monitors) reaches the CELL pin through a push-pull (SN74LVC1G17: actively high when ready, low otherwise: unpowered = loff, the card's 100 k pull-down reads not ready (INT-14).

FAIL-SAFE OFF: THE THREE LOSS CASES



1 CABLE UNPLUGGED: no +24V => no gate bias, no logic: every gate sits at 0 V on its 10k gate-source resistor (VGS(th) >= 1.9 V). With +24V still present, an open pair reads -0.30 V -> PWM and EN LOW.
2 CTRL UNPOWERED, or rev C card not enabling (it tri-states its PWM/EN drivers unless CHB_OK, its enable and our RDY are high, INT-15): driver outputs HI-Z (<= 100 uA, SLSB4BC p5) -> -0.29 V -> LOW. In reset the card drives 0 (10k on its ePWM balls, EN = GATE_EN AND CHB_OK AND MCU enable).
3 +3V3 ABSENT: LVC1A05 loff -> 10k pull-downs hold EN_DRV low -> AHCT; EN_GD, PWMs low: RDY_CELL HI-Z -> card reads 0 V: FLT HI-Z = fault, +5V ABSENT: NSI6651 VCC1 UVLO -> OUT low (p28), UCC14241-Q1 ENA = 0 -> bias off, RDY / DRV_FLT_N lose their pull-ups (to +5V) -> not ready / fault, +24V REMOVED (0-012 channel B): RDY low at >= 19.7 V -> EN_DRV and PWM low while bias regulates.

PWM GATING + 5 V INPUT LEVEL (FAST PATH)



PWMx = PWMx_RX AND EN_DRV, EN_GD = EN_DRV, both at +5V: the NSI6651 input thresholds are specified at VCC1 = 5 V only (VINH <= 3.5 V, p7), so VCC1 = +5V and the AHCT gates (VINH 2.0 V) lift the 3.3 V logic. A trip removes the commands within ~20 ns, ahead of the RST/EN filter (0.48-0.8 us). +5V or +3V3 absent, EN_DRV / PWMx pulled down, AHCT unpowered -> 0, NSI6651 VCC1 UVLO -> OUT low (functional modes p28).

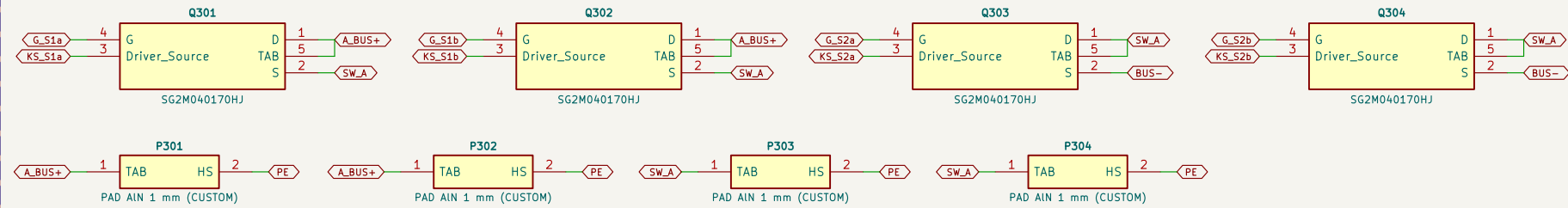
Not bench-validated. Values: sim/out/pv_design/cellSpec.json; checks: design_check()
Sheet 3
Schematics authored by Chinmoy Bhuyan
HV-PELV reinforced 1000 V DC (EC0-10); barriers listed on the cover

Sheet: /02 Fail-safe RS-422, fault latch/
File: 02_logic.kicad_sch

Title: PVCELL-25 power cell - Fail-safe RS-422, fault latch

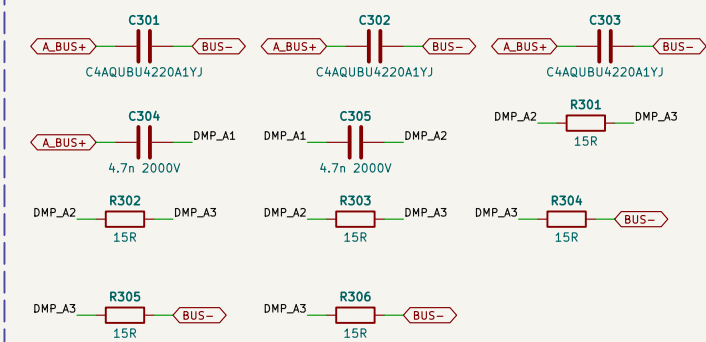
Size: A2 Date: 2026-10-04 Rev: C0
KiCad E.D.A. 10.0.6 Id: 3/10

LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



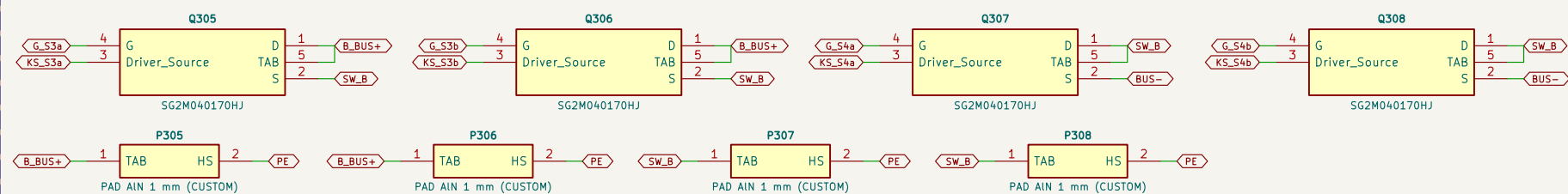
Kelvin source (pin 3) of each device returns to its own driver COM (gate-drive sheet).
Each tab on its own AIN ceramic (aluminum nitride) pad (P...: basic insulation HV-PE, IC-07; spec in the BOM). LAYOUT (IC-22):
milled slot between drain pin 1 and pins 2-4. SYMMETRY: equal drain/source busbar paths per switch.

LEG A DECOUPLING + RC DAMPER



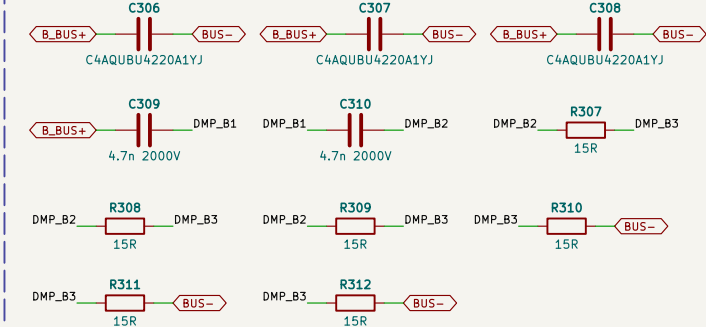
3 x C4AQUBU4220A1YJ at the S1/S2 pins: commutation loop cap-top-bottom ≤ 20 nH, bulk bank ≤ 15 nH from the leg (LAYOUT REQUIREMENTS, leg decd). Damper (PVR-02): 3.75 W normal / 5.68 W trip corner per leg into 2 x 3 CRCW2512-HP (7.4 W rated at 85 C, 500 V); damper loop at the device pins ≤ 3 nH.

LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



Kelvin source (pin 3) of each device returns to its own driver COM (gate-drive sheet).
Each tab on its own AlN ceramic (aluminium nitride) pad (P...: basic insulation HV-PE, IC-07; spec in the BOM). LAYOUT (IC-22):
milled slot between drain pin 1 and pins 2-4. SYMMETRY: equal drain/source busbar paths per switch.

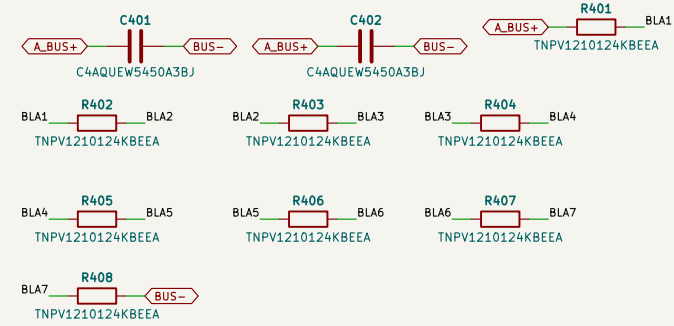
LEG B DECOUPLING + RC DAMPER



3 x C4Q4UBU4220A1YJ at the S3/S4 pins: commutation loop cap-top-bottom ≤ 20 nH, bulk bank ≤ 15 nH from the leg (LAYOUT REQUIREMENTS, leg deck). Damper (PVR-02): 3.75 W normal / 5.68 W trip corner per leg into 2 x 3 CRCW2512-HP (7.4 W rated at 85 C, 500 V): damper loop at the device pins ≤ 3 nH.

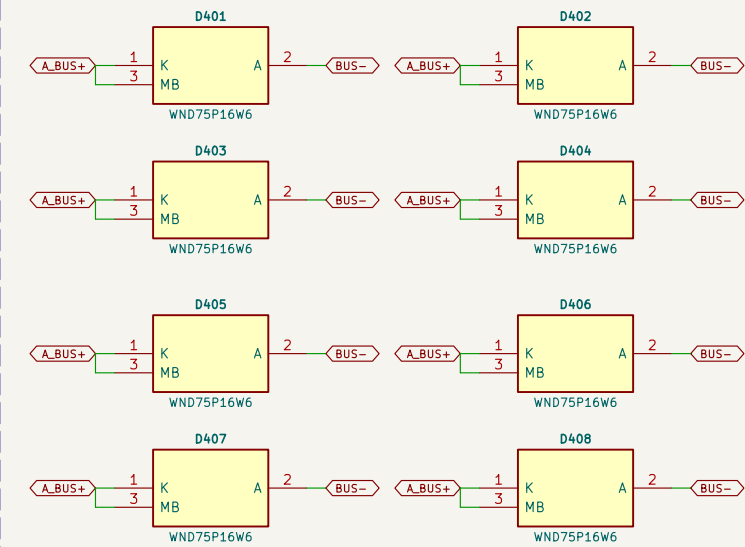
Not bench-validated. Values: sim/out/pv_design/cell_spec.json; checks: design_check() Sheet 4 Schematics authored by Chinmoy Bhuyan HV-PELV reinforced 1000 V DC (ECO-10); barriers listed on the cover	
Sheet: /03 Power stage, decoupling, dampers/ File: 03_power.kicad_sch	
Title: PVCELL-25 power cell – Power stage, decoupling, dampers	
Size: A3	Date: 2026-10-04
KiCad E.D.A. 10.0.6	Rev: C0 Id: 4/10

PORT A FILM BANK + BLEEDER



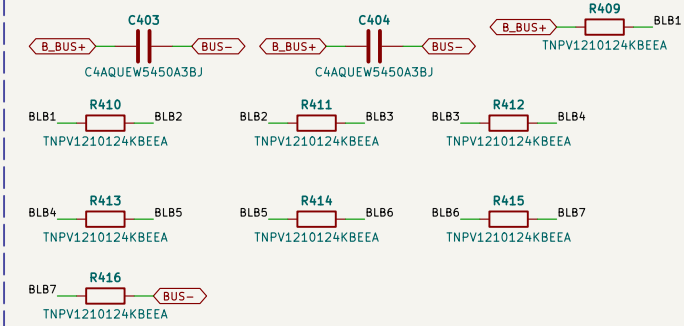
2 x C4AQUEW5450A3BJ = 90 uF, ripple 23.6 of 47.6 A rms (use limit). Bleeder 8 x TNPV1210 124k: 138 V and 152 mW per element at 1100 V; bank to 60 V in <= 5 min. Busbar to the leg <= 15 nH.

PORT A BANK REVERSE CLAMP (8 X WND75P16W6)



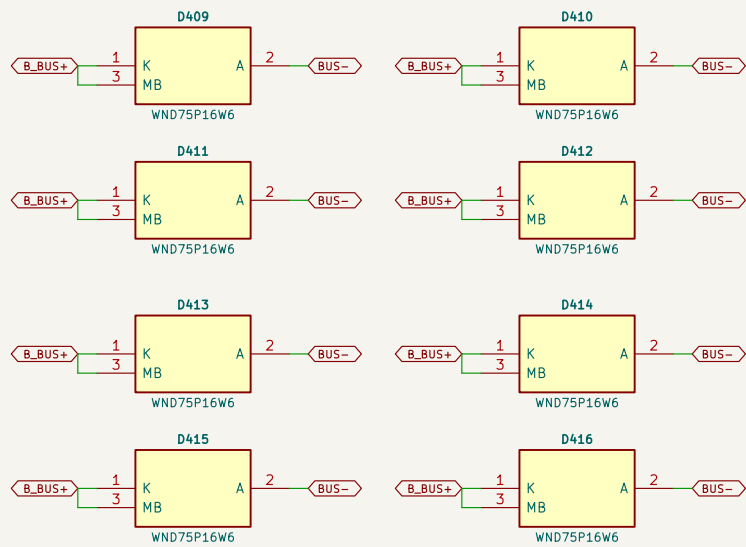
A bolted terminal short rings the bank below zero; these carry the freewheel instead of the SiC body diodes. Cathode to A_BUS+, anode to BUS-, in parallel. LAYOUT REQUIREMENT: on the laminated bus AT the C4AQ terminals, spread symmetrically between the capacitors, bank $\rightarrow$ group $\rightarrow$ bank ≤ 1.5 nH.

PORT B FILM BANK + BLEEDER



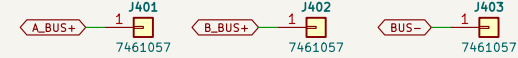
2 x C4AQUEW5450A3BJ = 90 uF, ripple 23.6 of 47.6 A rms (use limit). Bleeder 8 x TNPV1210 124k: 138 V and 152 mW per element at 1100 V; bank to 60 V in <= 5 min. Busbar to the leg <= 15 nH.

PORT B BANK REVERSE CLAMP (8 X WND75P16W6)



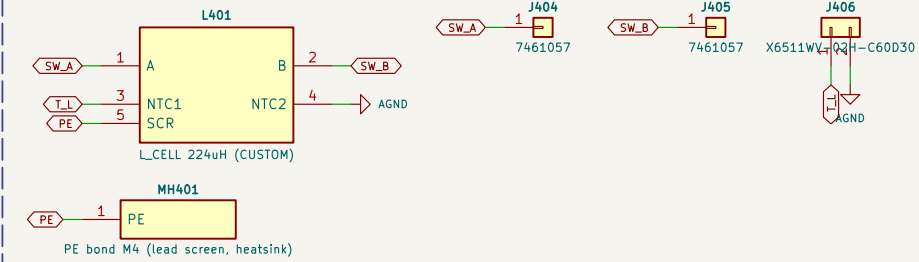
A bolted terminal short rings the bank below zero; these carry the freewheel instead of the SIC body diodes. Cathode to B_BUS+, anode to BUS-, in parallel. LAYOUT REQUIREMENT: on the laminated bus AT the C4AQ terminals, spread symmetrically between the capacitors, bank $\rightarrow$ group $\rightarrow$ bank ≤ 1.5 nH.

BUS TERMINALS TO PV-PORT



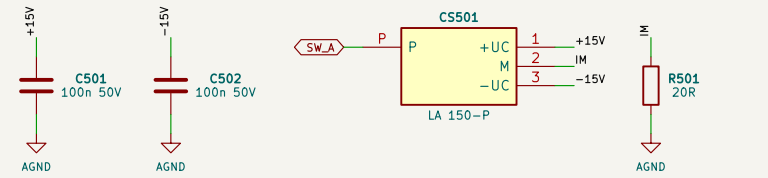
REDCUBE M3 studs, 100 A each: A_BUS+, B_BUS+, BUS- (net names as gen/port.py).

INDUCTOR (OFF-BOARD CUSTOM) + TERMINALS



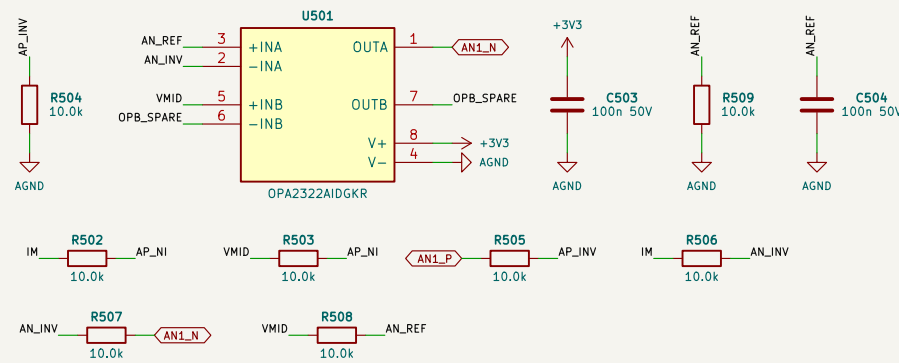
226 uH at 45 A (L0 244 uH), 37 turns on NPC290026 x 2; full spec in the part description / BOM.
Studs SW_A (lead A through the LA 150-P) and SW_B. Winding NTC (reinforced to the winding) on J.

AN1: INDUCTOR CURRENT (LEM LA 150–P AROUND INDUCTOR LEAD A)



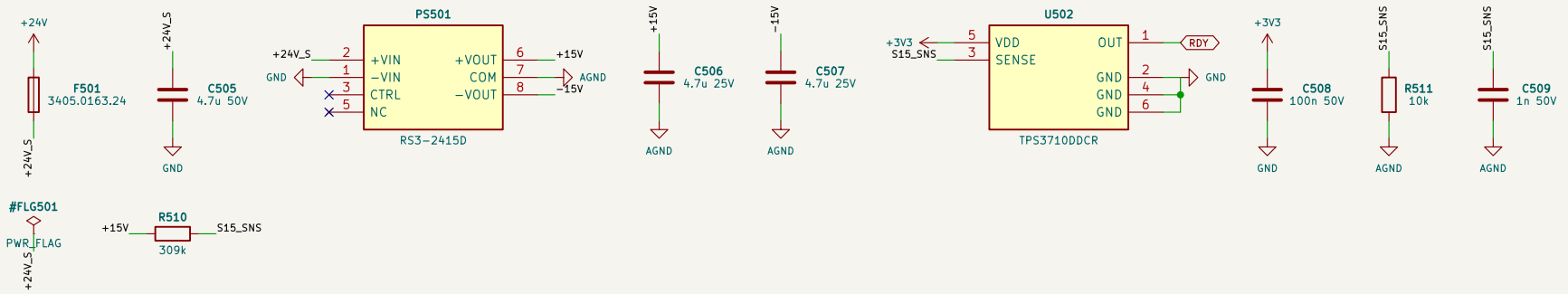
Lead A of L_CELL (Insulated, screen earthed in the aperture) passes through the transducer; positive = SW_A -> inductor -> SW_B. IS = IP/2000 into RM 20R to AGND; 10 mV/A. Linear +/-11V A delay <= 0.52 us; CTRL backup peak 106.0 A (< 129 A). LAYOUT: lead <= 100 C at the transducer (LA 150–P p4). RM and the +/-15 V return at one AGND point next to the transducer; no other conductor through the aperture.

AN1 DIFFERENTIAL DRIVER (20 mV/A, CM 1.25 V)



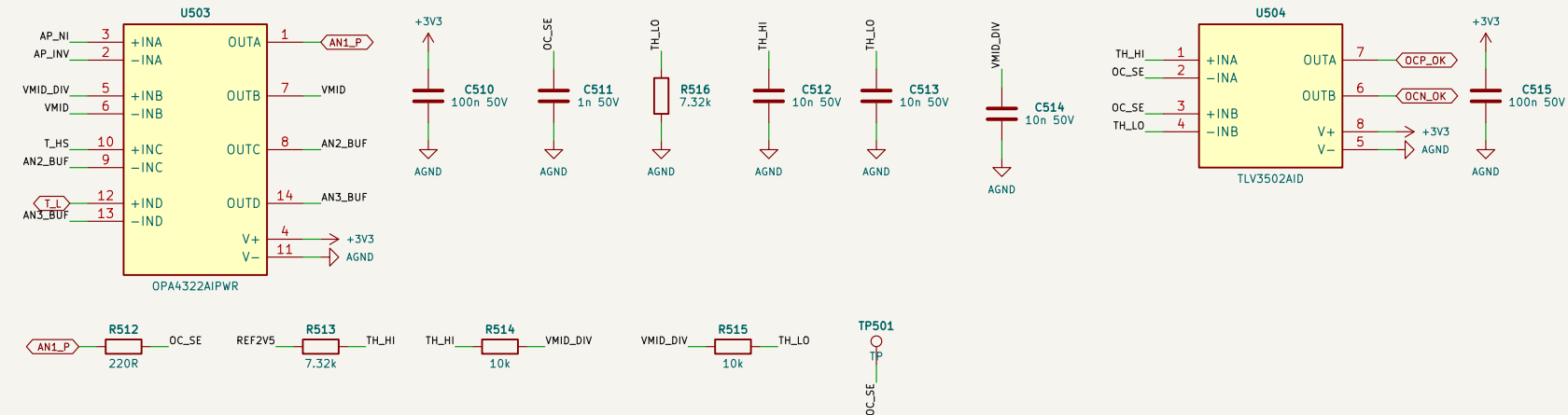
OPA4322 A: AN1_P = VMID + V_M (gain 2 on (V_M + VMID)/2); OPA2322 A: AN1_N = VMID - V_M (inverting, + at VMID/2). Direct drive (< 1 ohm closed loop) of the ribbon and the CTRL 1.50 k open-input bias (<= 2.3 mA). CTRL WFE 0.402 -> 0.0759 A/LSB. An unplugged cable reads far negative on CTRL (its bias).

+/-15 V FOR THE LA 150–P (RS3–2415D)



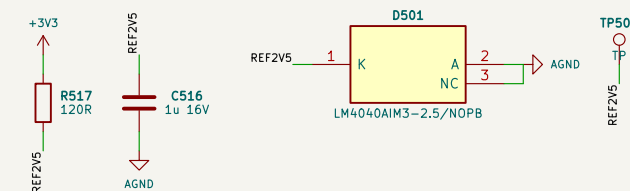
From +24V through a 0.5 A quick fuse (RECOM requires one); isolation of the module not used; COM = AGND. TPS3710 holds RDY low below 12.5–12.8 V on +15 V; a dead sensor supply never reads as 0 A while enabled. -15 V is not monitored separately (one converter); firmware checks AN1 plausibility.

OC WINDOW COMPARATOR +/-72.4 A (BOTH POLARITIES)



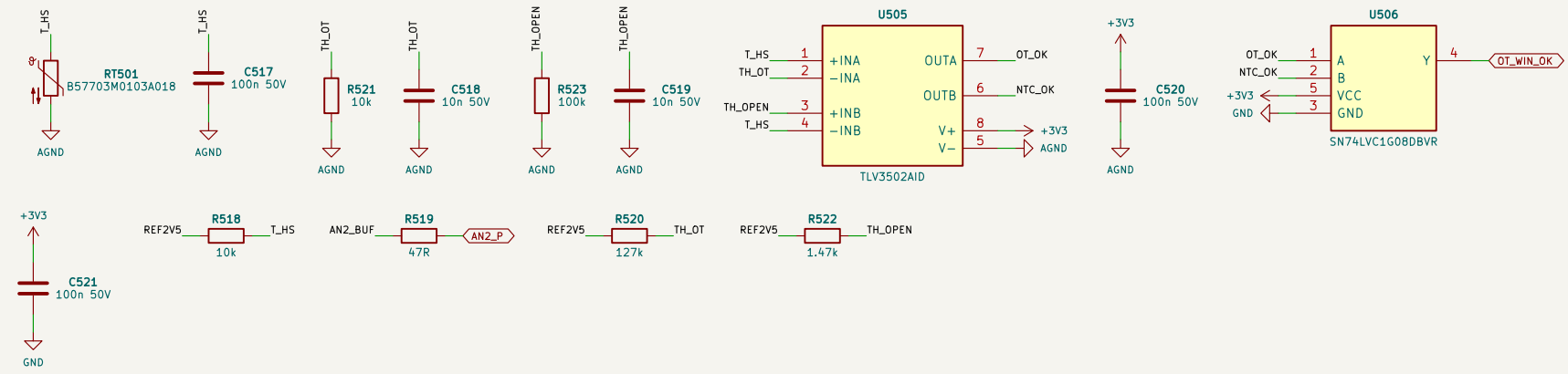
TLV3502 on AN1_P through 220R/1n; OCP_OK low above TH_HI, OCN_OK low below TH_LO -> latch. Trip 68.5–76.0 A (all tolerances); gate off 0.94 us after the crossing (PWM path), 1.73 us via EN; allowed 3.66 us. OPA4322 B: VMID buffer for the AN1 driver.

2.5 V REFERENCE



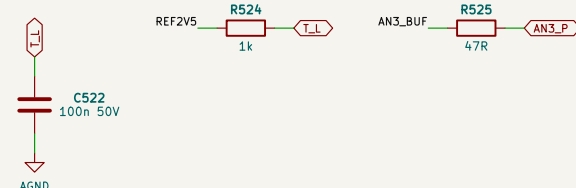
LM4040 A grade from 3V3 through 120R; 2.6 mA worst load + 65 uA, <= 8.4 mA total (15 mA max). Feeds the OC thresholds, the OT threshold and both NTC dividers (OT is reliometric).

AN2 HEATSINK TEMPERATURE + OT TRIP 95 C



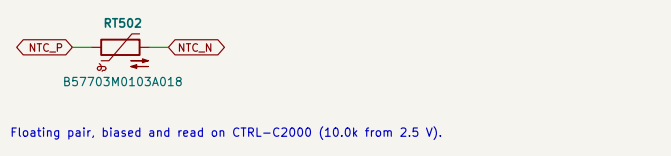
2.5 V - 10k - T_HS - NTC (B57703M, screwed next to the hottest device) - AGND; 2.43 V at -40 C ... 0.045 V at 150 C; buffered (OPA4322 C) to AN2_P via 47R. TLV3502 A: low above 95.0 C (92.3–97.9 C); B: low when T_HS > TH_OPEN (open probe reads 2.5 V; -40 C reads 2.432 V) -> OT_WIN_OK = OT_OK AND NTC_OK -> latch (PVR–08).

AN3 INDUCTOR WINDING TEMPERATURE



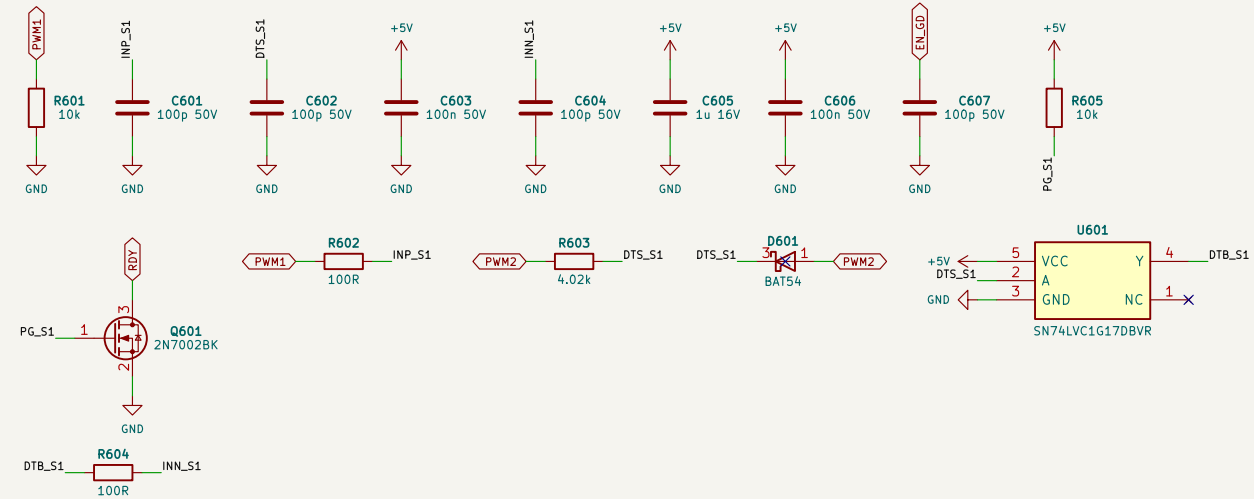
2.5 V - 1k - T_L - winding NTC (inside L_CELL header on sheet 04) - AGND; the NTC lead next to the 124 V/ns winding ends on AGND and the 100 nF T_L node; never on the reference. Buffered (OPA4322 D) to AN3_P via 47R; monitoring only (no local trip in the spec). Open NTC reads 2.5 V like -40 C; firmware plausibility check (winding colder than the heatsink at load = fault). Fan law and hot-spot limit in firmware.

NTC PAIR: SECOND HEATSINK NTC



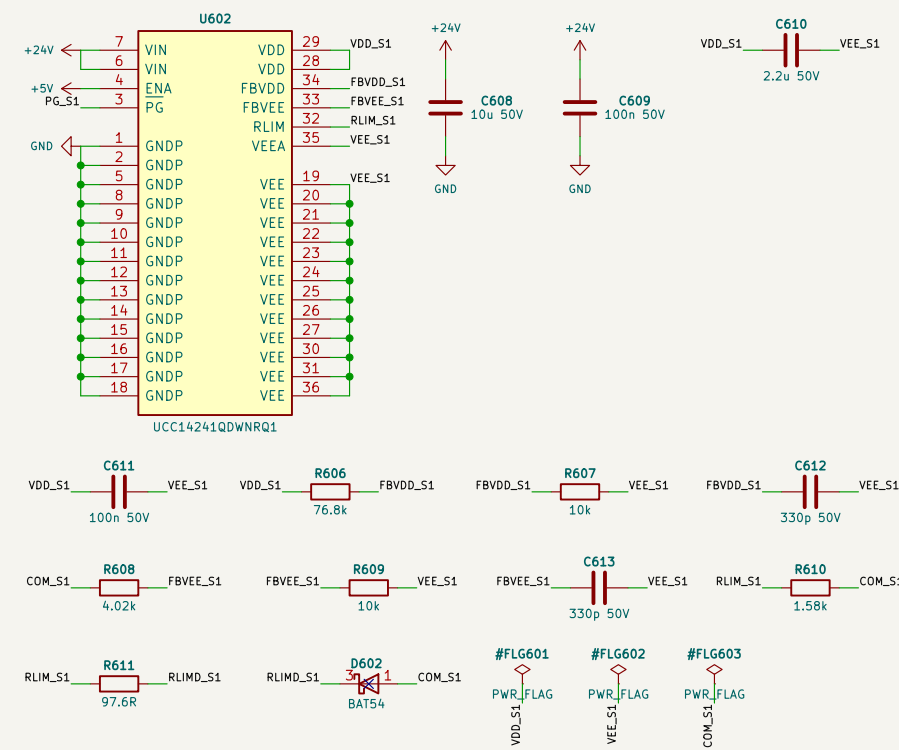
Floating pair, biased and read on CTRL–C2000 (10.0k from 2.5 V).

S1: LOGIC SIDE



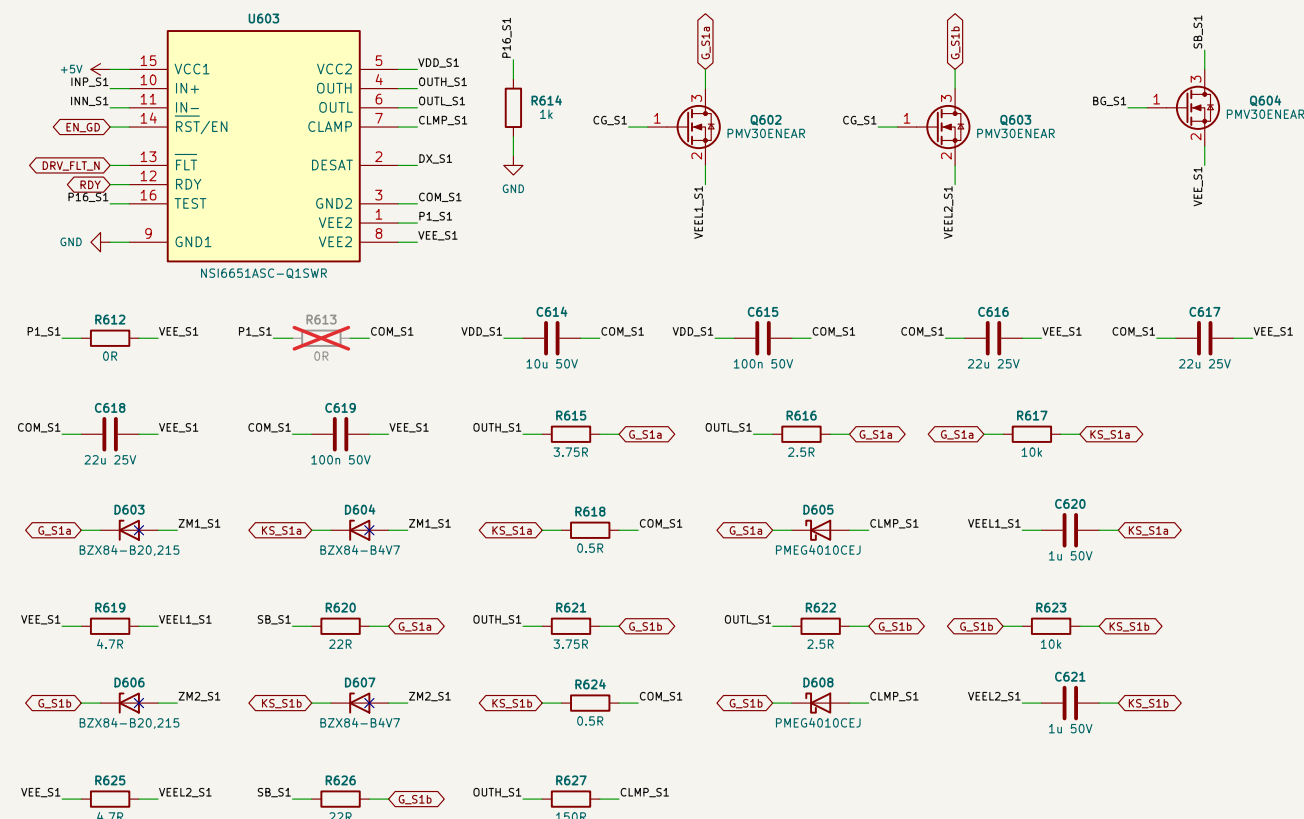
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

S1: ISOLATED BIAS +18 V / -3.5 V



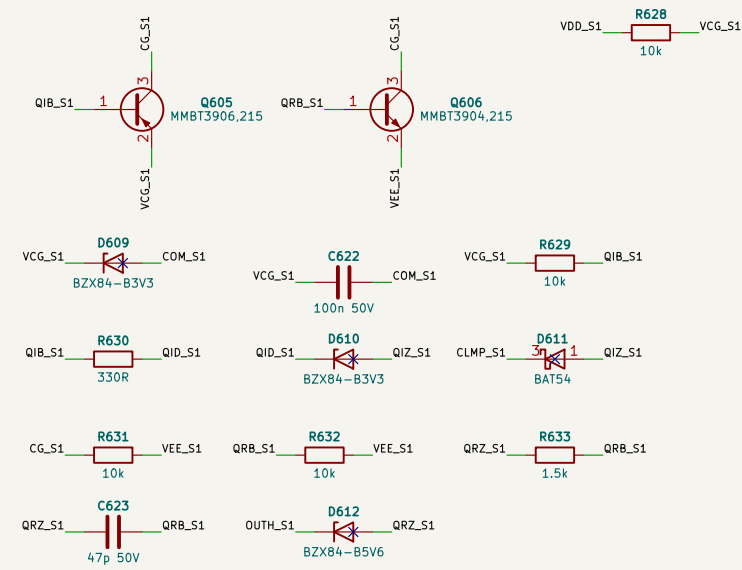
UCC14241-Q1, 24 V in, regulated +/-1.3 %, reinforced barrier.
RDR: RLIM1 sources COM, RLIM2 + DLIM sink it.

S1: DRIVER AND 2 GATES



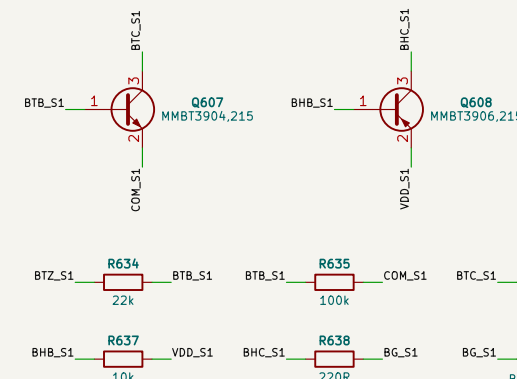
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold; N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

S1: MILLER CLAMP CONTROL



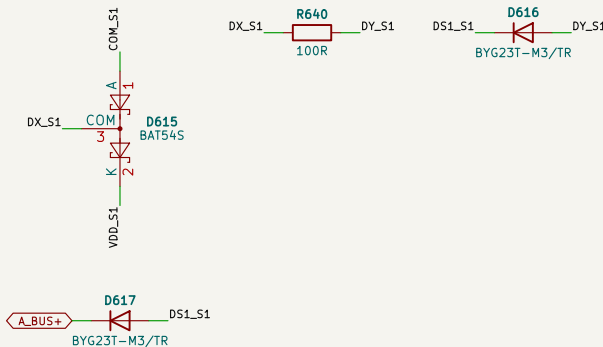
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

S1: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V;
per device AD3400A + 22R pull the gate to VEE; BG holds -3 uA (1k + FET gates).

S1: DESAT



DESAT pin (internal 500 uA blanking source, trip 8.5-10.0 V) - Rs - 2 x BYG231-M3/TR (1300 V each) - drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; CoK footprint DNP.

S1: LAYOUT RULES + MILLER MARGIN

SYMMETRY: gate and Kelvin of S1a/S1b as coupled pairs of equal length (+/-10 %) from the COM_S1 star; Ron.
Ruff, Kelvin R, 10k; Zeners at each device; power sources joined only by the busbar; CLAMP LOOP: each
AD3400A + C_VL at its device's gate-Kelvin pins, loop <= 1 nH, IC-22; milled slot between the drain
pin 1 and pins 2-4 of each 10-24V-4L; DESAT string >= 2.5 mm pad gap per diode; board coated to IPD1.
Miller hold (gdr rev 5, ngpsdc 1100 V / 72A A, 115 V/mpl; die +1.41 V + VGS(0)) min +9k V at 5.75 C,
margin 0.53 V (rev 4 PNP; +2.14 V) - BENCH TEST (double pulse, hot, minimum-threshold devices).

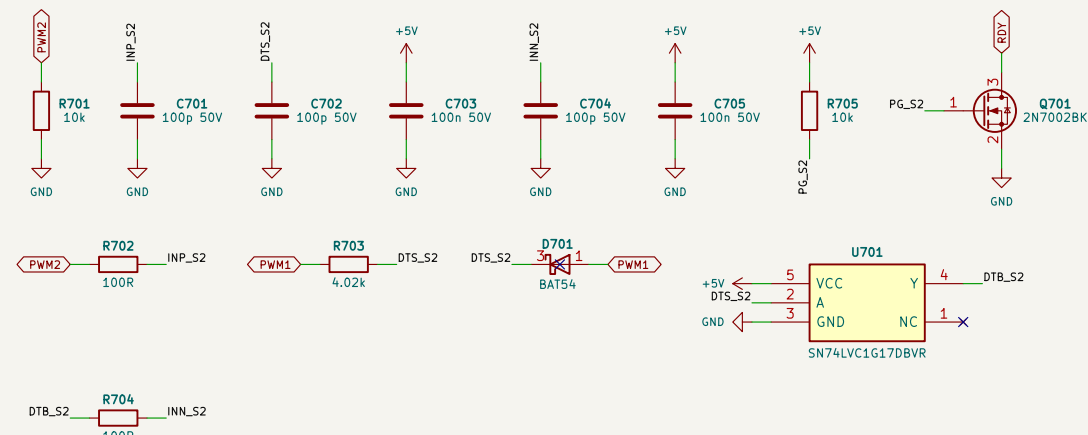
Not bench-validated. Values: sim/out/pw_design/cellSpec.json; checks: design_check()
Sheet 7
Schematics authored by Chinmoy Bhuyan
HV-PELV reinforced 1000 V DC (ECO-10); barriers listed on the cover

Sheet: /06 Gate drive S1 (leg A)/
File: 06_gd_s1.kicad_sch

Title: PVCELL-25 power cell - Gate drive S1 (leg A)

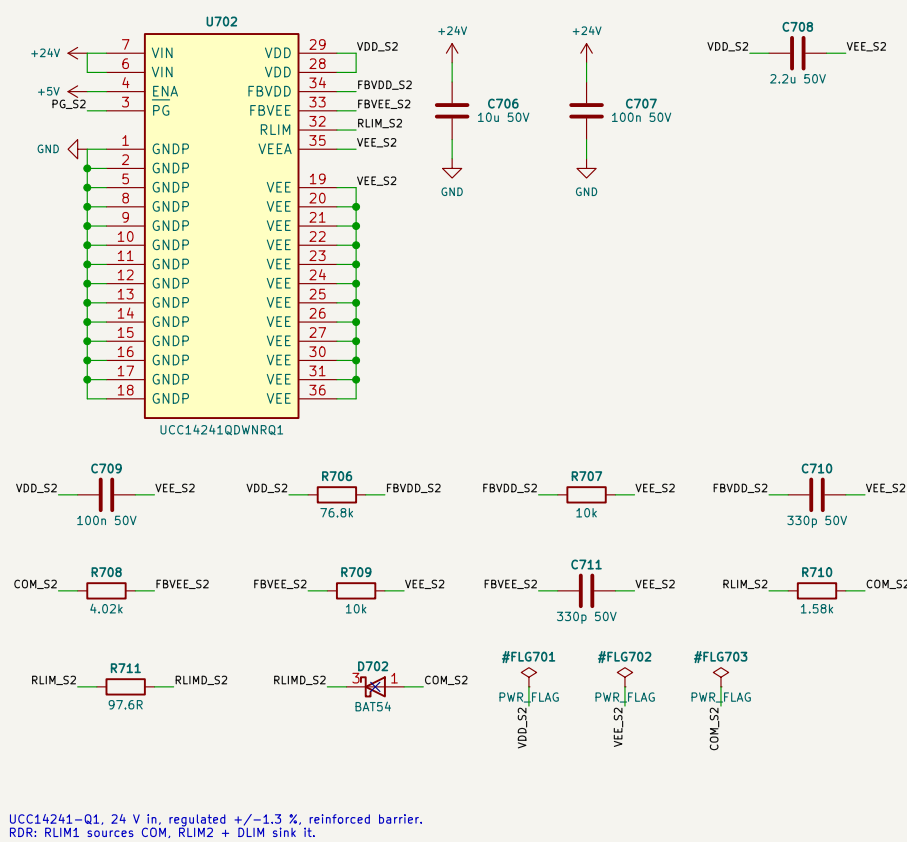
Size: A2 Date: 2026-10-04 Rev: C0
KiCad E.D.A. 10.0.6 Id: 7/10

S2: LOGIC SIDE



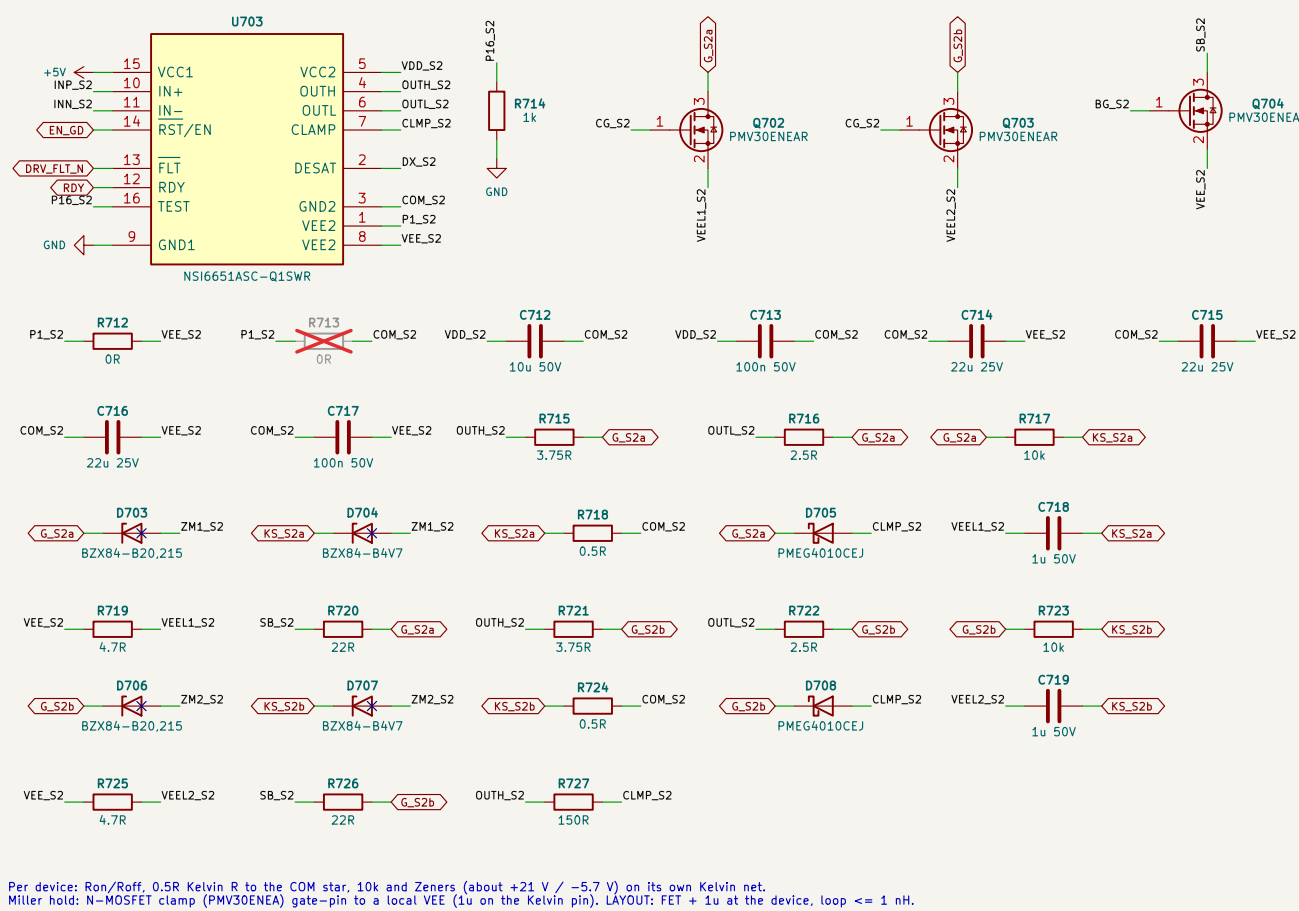
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

S2: ISOLATED BIAS +18 V / -3.5 V



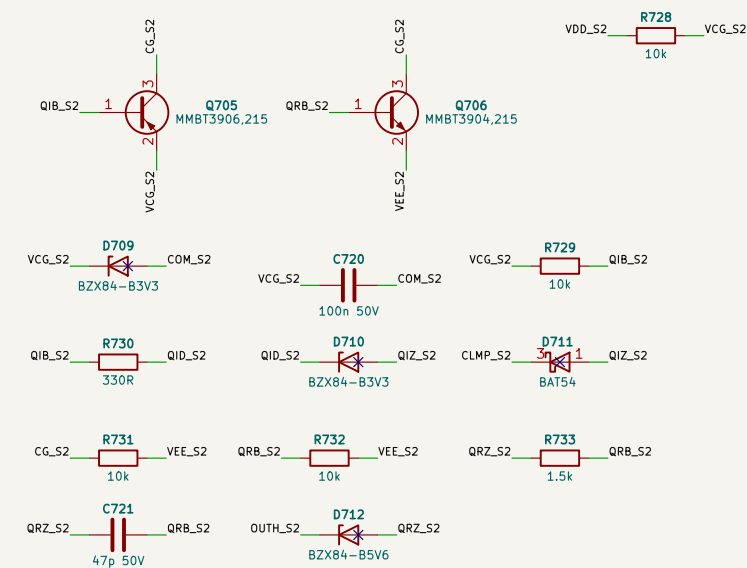
UCC14241-Q1, 24 V in, regulated $\pm 1.3\%$, reinforced barrier.
RDR: RLIM1 sources COM, RLIM2 + DLIM sink it.

S2: DRIVER AND 2 GATES



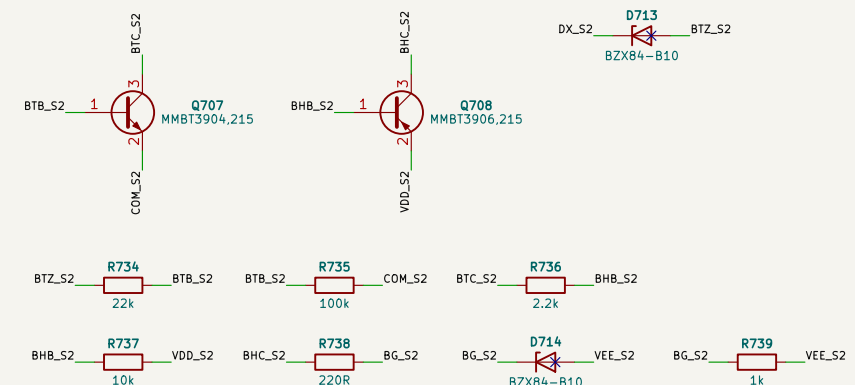
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

S2: MILLER CLAMP CONTROL



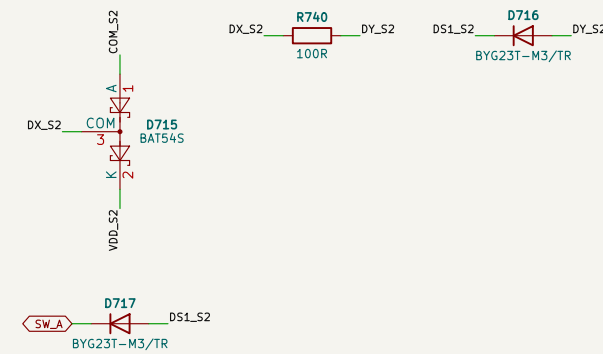
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V.
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

S2: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device A03400A + 22R pull the gate to VEE; BG holds -3 us (1k + FET gates).

S2: DESAT



DESAT pin (internal 500 uA blanking source, trip 8.5–10.0 V) – Rs – 2 x BYG23T-M3/TR (1300 V each) – drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

S2: LAYOUT RULES + MILLER MARGIN

SYMMETRY: gate and Kelvin of S2a/S2b as coupled pairs of equal length (+/-10 %) from the COM.S2 star; Ron, Roff, Kelvin = 10k; Zenith = 10k; each device coupled source joined only by the busbar. CLAMP LOOP: e.g. A03A00A + C-VL at its device's gate-Kelvin pins, loop = 1 nC, 1H; IC-12: milled slot between the drain pin 1 and pins 2-4 of device TO-247-4AL; DESAT string = 2.5 mm pad gap per diode; board coated to Pd1. Miller hold (gdrv rev 5, ngpsice 1100 V / 72.4 A, 115 V/ns) die +1.41 V vs VG5(th) min 1.94 V at 175 °C, margin 0.53 V (rdv 4P VNP; +2.14 V) - BENCH TEST (double pulse, hot, minimum-threshold devices).

Not bench-validated. Values: sim/out/pv_design/cell_spec.json; checks: design_check()
Sheet 8
Schematics authored by Chinmoy Bhuyan
HV-PELV reinforced 1000 V DC (ECO-10); barriers listed on the cover

Sheet: /07 Gate drive S2 (leg A)/
File: 07_gd_s2.kicad_sch

Title: PVCELL-25 power cell – Gate drive S2 (leg A)

Size: A2 Date: 2026-10-04

KiCad E.D.A. 10.0.6	Id: 8/10
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10	11
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The schematic diagram illustrates the power supply section of the 74100B8K. It features a multi-tap transformer (R801, 10k) connected to a bridge rectifier (D801, BAT54) with a 5V output. The rectifier output is connected to a series of capacitors (C802, C803, C804, C805, C806) and a resistor (R805, 10k) to a 5V output. The output is connected to a 2N7000B8K MOSFET (Q801) which drives the gate of a 74100B8K MOSFET (U801). The 74100B8K MOSFET is connected to a 5V output and a 100k resistor (R804) to the 5V output.

[illegible]

The schematic diagram illustrates the internal and external components of the NS16651ASC-Q1SWR circuit. The internal block diagram shows the connection of various pins to the external circuit. The external component layout shows the placement of components like resistors, capacitors, and diodes on the PCB.

Internal Block Diagram:

- U003:** A central block representing the NS16651ASC-Q1SWR. It has several pins connected to external components:
 - VCC1 (15):** Connected to +5V.
 - INP_S3 (10):** Connected to INN_S3 (14).
 - IN+ (11):** Connected to IN- (12).
 - IN- (12):** Connected to IN+ (11).
 - RST/EN (7):** Connected to EN_CD (14).
 - FLT (13):** Connected to DRV FLT N (12).
 - RDY (12):** Connected to RST/EN (7).
 - TEST (16):** Connected to RST/EN (7).
 - GND1 (9):** Connected to GND.
 - VCC2 (5):** Connected to VDD_S3 (5).
 - OUTH (4):** Connected to OUTH_S3 (4).
 - OUTL (6):** Connected to OUTL_S3 (6).
 - CLAMP (3):** Connected to CLMP_S3 (3).
 - DESAT (2):** Connected to DX_S3 (2).
 - COM_S3 (3):** Connected to COM_S3 (3).
 - P1_S3 (1):** Connected to P1_S3 (1).
 - VEE2 (8):** Connected to VEE_S3 (8).

External Component Layout:

- Power and Ground:**
 - +5V:** Connected to VCC1 (15).
 - GND:** Connected to GND1 (9).
- Signal and Control:**
 - EN_CD (14):** Connected to EN_CD (14).
 - DRV FLT N (12):** Connected to DRV FLT N (12).
 - RST/EN (7):** Connected to RST/EN (7).
- Power and Ground:**
 - VDD_S3 (5):** Connected to VDD_S3 (5).
 - VEE_S3 (8):** Connected to VEE_S3 (8).
- Signal and Control:**
 - OUTH_S3 (4):** Connected to OUTH_S3 (4).
 - OUTL_S3 (6):** Connected to OUTL_S3 (6).
 - CLMP_S3 (3):** Connected to CLMP_S3 (3).
 - DX_S3 (2):** Connected to DX_S3 (2).
 - COM_S3 (3):** Connected to COM_S3 (3).
 - P1_S3 (1):** Connected to P1_S3 (1).

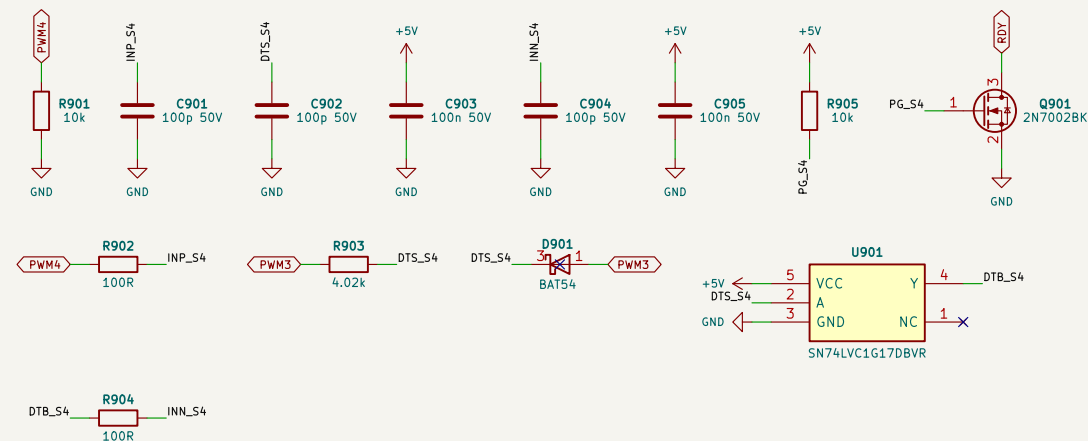
[illegible]

Figure 1: Schematic representation of the genetic organization of the BTZ-33 locus. The figure shows two panels, (A) and (B), representing different alleles. Panel (A) shows the Q807 allele with a 1.2 kb region containing BTZ_S3, Q807, and COM_S3. Panel (B) shows the Q808 allele with a 1.2 kb region containing BHB_S3, Q808, and VOD_S3. Both panels show the presence of DX_S3 and BTZ_S3 regions. The figure also includes a legend for the R334, R335, R336, R337, R338, R339, and R340 regions, which are represented by red boxes with numbers 22k, 100k, 22k, 10k, 220k, and 1k respectively.

[illegible]

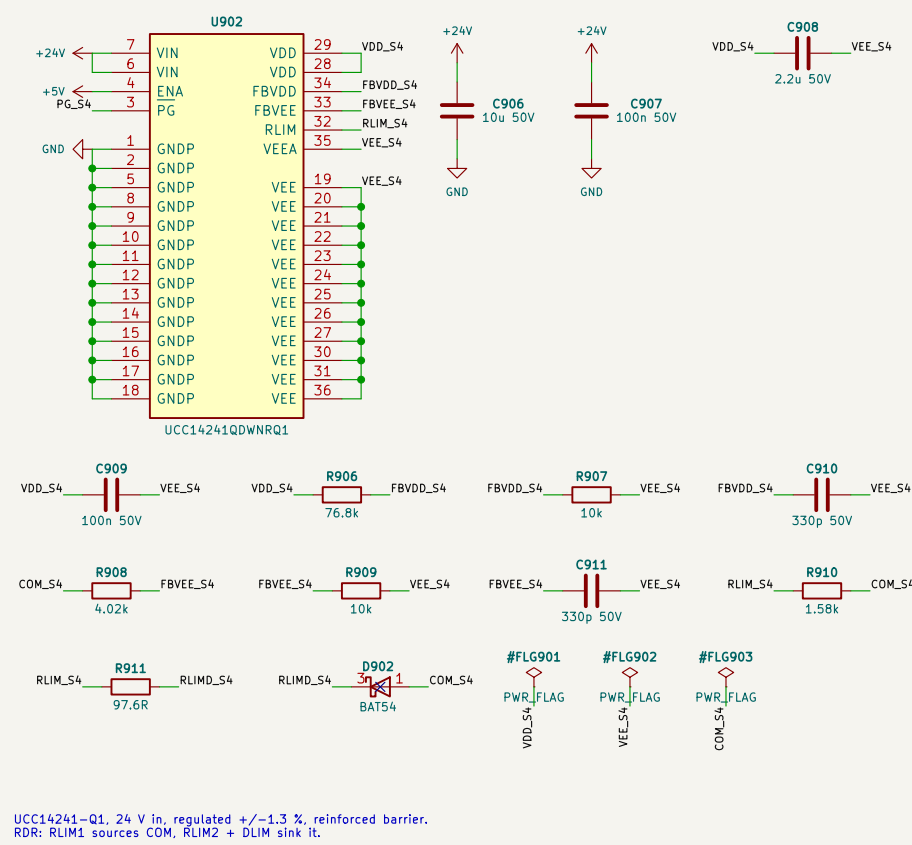
SYMMETRY: gate and Kelvin of S3a/S3b as coupled pairs of equal length (+/-10 %) from the COM.S3 star; Ronf, Ref. Kelvin R. 30k; Zenins at each device; power sources joined only by the busbar. CLAMP LOOP: each ADS300A (1.7 g) at its device's gate-Kelvin pins, (iso ≤ 1 m, IC ≥ 22 milled slot between the drain pin and pins 2-4 of the D0-247-4-L) DESAT string = 0.5 mm pad gap per device; board coated to PDI. Miller hold (drv rev 5, nupic 1500 V / 72.4 A, 155 V/s); die +2.14 V vs VGS(min) min -1.90 V at 175 C, margin 0.15 V (rev 4 PNP; +2.14 V) - bench test (double pulse, hot, minimum-threshold devices).

S4: LOGIC SIDE



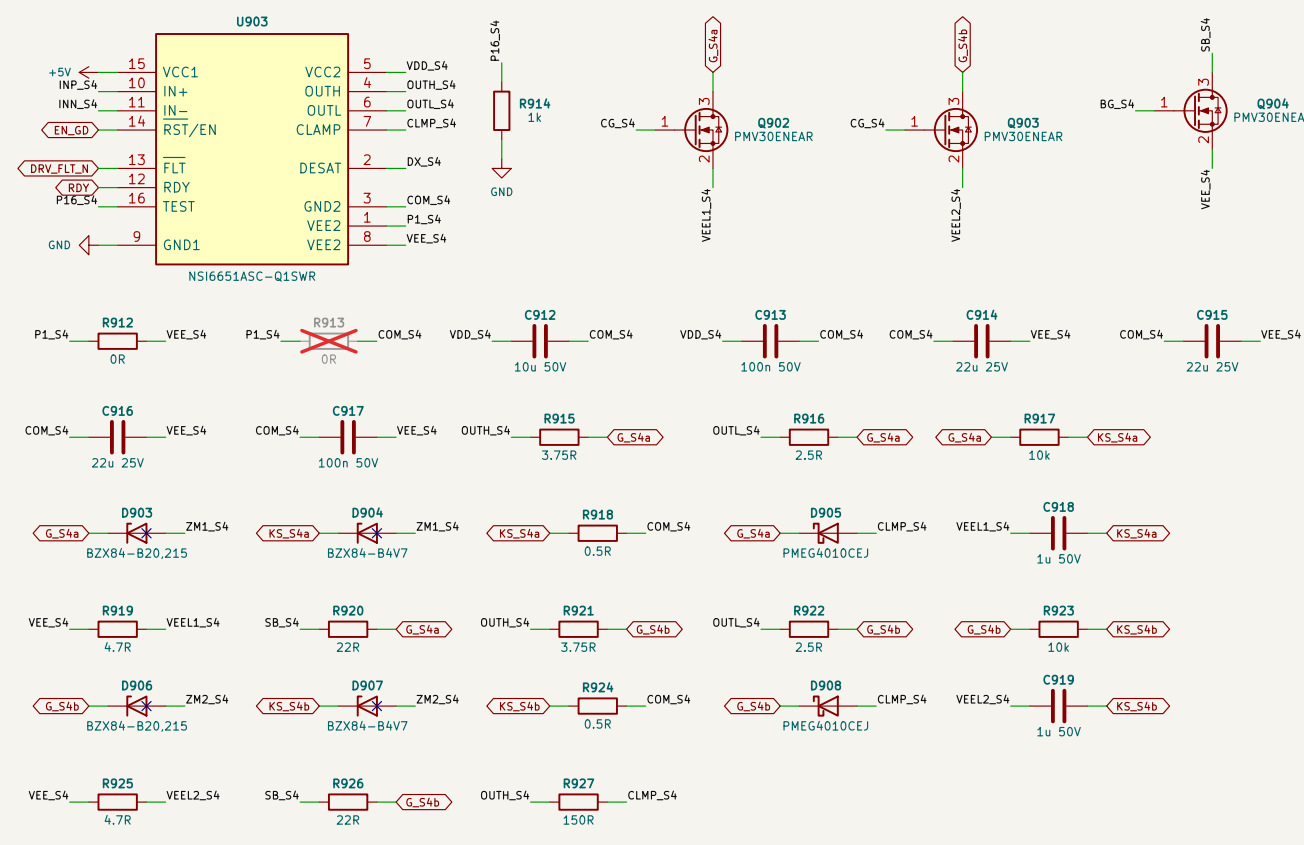
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

S4: ISOLATED BIAS +18 V / -3.5 V



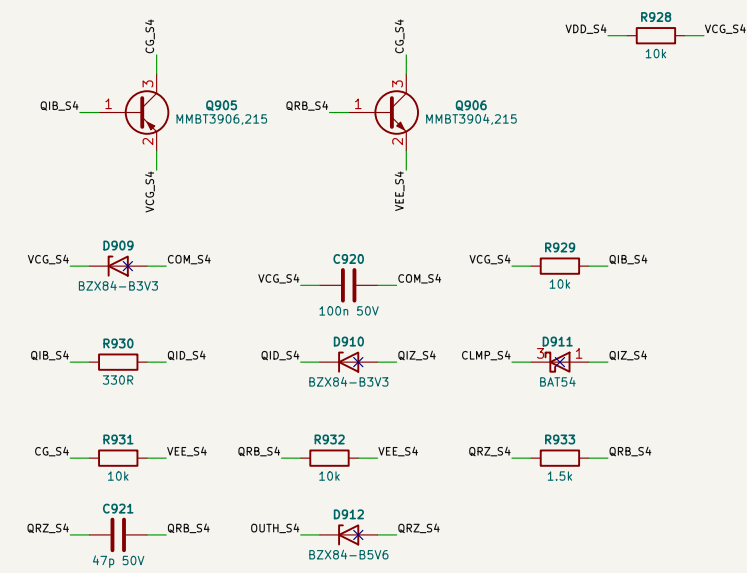
UCC14241-Q1, 24 V in, regulated $\pm 1.3\%$, reinforced barrier.
RDR: RLIM1 sources COM, RLIM2 + DLIM sink it.

S4: DRIVER AND 2 GATES



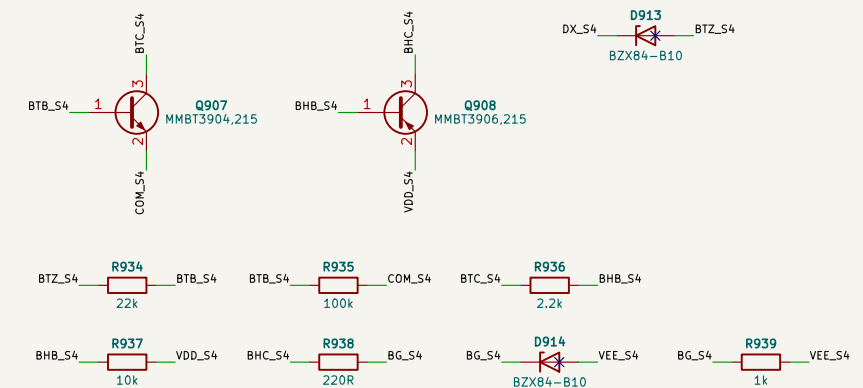
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30NEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

S4: MILLER CLAMP CONTROL



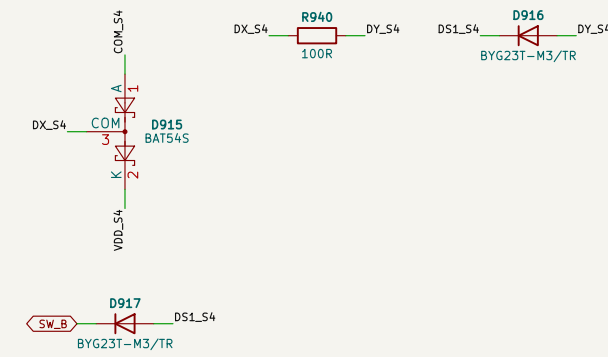
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

S4: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 10.0 V maximum trip of the -Q1 grade) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device A03400A + 22R pull the gate to VEE; BG holds -3 us (1k + FET gates).

S4: DESAT



DESAT pin (internal 500 uA blanking source, trip 8.5–10.0 V) – Rs – 2 x BYG23T-M3/TR (1300 V each) – drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cblk footprint DNP.

S4: LAYOUT RULES + MILLER MARGIN

SYMMETRY: gate and Kelvin of S4a/S4b as coupled pair of equal length (+/-10 %) from the COM_S4 star; Ron, Rgv, Kelvin R, 10k, Zeners at each device; power sources joined only by the busbar. CLAMP LOOP: each A03400A + C_VL at its device's gate-Kelvin pins, loop <= 1 nH. IC12:2: milled slot between the drain pin 1 and pins 2-4 of each T0-247-4L; DESAT string = 2.5 mm pad gap per diode; board coated to PD1. Miller hold (gdrv rev 5, npsicpe 1100 V / 72.4 A, 115 V/ns); die +1.41 V vs VGs(th) min 1.94 V at 175 °C, margin 0.53 V (rev 4) +2.14 V) – BENCH TEST (double pulse, min. minimum-threshold devices).